



**Department of Information and Communication Technology
Faculty of Science and Technology, Bangladesh University of Professionals**

THESIS PROPOSAL

Deploying Distilled AI Models for Efficient and Real-Time Email Phishing Detection

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APPROVAL

The project titled Deploying Distilled AI Models for Efficient and Real-Time Email Phishing Detection submitted by Nasrat Jahan, Md. Rasheduzzaman Imran, Md. Murad Jahan of Session 2021-22 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Bachelor of Science in Information and Communication Engineering under department of Information and Communication Technology on February 2026.

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Abstract

Email phishing remains a persistent and evolving cybersecurity threat, causing significant financial losses for organizations worldwide. While transformer-based models such as BERT have demonstrated high accuracy in detecting phishing content, their substantial computational requirements hinder real-time deployment in enterprise environments. This research proposes a comprehensive framework for deploying knowledge-distilled AI models to enable efficient, real-time email phishing detection. The methodology involves optimizing knowledge distillation techniques to transfer detection capabilities from a large BERT teacher model to a compact TinyBERT student architecture. The distilled model is integrated with traditional cybersecurity feature engineering and deployed as a scalable containerized microservice using FastAPI and Docker. The system is designed to process high-volume email traffic with low latency while maintaining enterprise-grade security standards. Expected outcomes include a production-ready framework achieving over 95% detection accuracy, with an 85% reduction in model size and a 60% improvement in inference speed compared to conventional BERT models. The research bridges the gap between advanced AI capabilities and practical deployment constraints, offering a scalable solution for organizations seeking robust, real-time phishing detection without prohibitive infrastructure costs. The findings will contribute to the fields of cybersecurity and natural language processing, with open-source resources made available for community adoption and further innovation.

Keywords: Phishing Detection, Knowledge Distillation, BERT, Model Deployment, Cybersecurity

1 Introduction

Email phishing remains one of the most significant and evolving cybersecurity threats in the digital landscape, with attackers continuously developing sophisticated social engineering tactics. The financial impacts of successful phishing attacks cost organizations billions annually. While large pre-trained language models like BERT demonstrate superior accuracy in identifying phishing content, their substantial computational requirements make them impractical for real-time email filtering in resource-constrained environments. This research addresses this critical challenge by proposing a knowledge-distilled AI framework that maintains high detection accuracy while meeting real-time processing requirements in enterprise settings.

The landscape of phishing detection has evolved considerably over the past two decades. Early detection methods relied primarily on list-based approaches, including blacklists and whitelists of URLs and domains, as well as heuristic rules that flagged emails based on predefined patterns. While computationally efficient, these methods suffer from fundamental limitations: they cannot detect zero-day threats, require constant manual updates, and fail to adapt to evolving attack strategies. The advent of machine learning introduced more sophisticated approaches, utilizing hand-crafted features extracted from email headers, URL structures, and content to train classifiers. More recently, the emergence of transformer-based architectures, particularly Bidirectional Encoder Representations from Transformers (BERT), has revolutionized natural language processing capabilities. These models demonstrate remarkable proficiency in understanding contextual relationships, semantic nuances, and linguistic patterns within text. In the context of phishing detection, BERT and similar models can analyze email content with near-human levels of comprehension, identifying subtle linguistic cues, deceptive patterns, and contextual anomalies that characterize sophisticated phishing attempts.

Despite these advances, the deployment of large transformer models in production environments presents substantial challenges. Standard BERT models contain approximately 110 million parameters, require significant memory resources, and exhibit inference latencies that make them impractical for real-time email filtering in high-volume enterprise environments. Production email systems at medium to large enterprises typically process thousands of messages per minute, requiring inference times measured in milliseconds rather than seconds. The hardware requirements for running full-scale transformer models impose costs that many organizations cannot justify, creating a situation where security teams must choose between high-accuracy but impractical models or deployable but less effective solutions.

The significance of this research lies in its potential to bridge the gap between state-of-the-art AI capabilities and practical deployment requirements. The increasing sophistication of phishing attacks demands advanced detection methods, but current solutions either sacrifice accuracy for efficiency or incur prohibitive computational costs. By developing a knowledge-distilled system that maintains high accuracy while being computationally efficient, this research provides a practical solution that can be widely adopted by organizations of various

sizes, ultimately enhancing overall cybersecurity resilience. The proposed framework represents a significant step toward making advanced deep learning models accessible for real-time security applications in resource-constrained environments.

This research proposes a comprehensive framework addressing the accuracy-efficiency trade-off through optimized knowledge distillation techniques. The methodology involves fine-tuning a BERT-based teacher model on phishing detection tasks, then distilling its capabilities into a compact TinyBERT architecture that maintains semantic understanding while dramatically reducing parameter count. The distilled model will be integrated with traditional cybersecurity feature engineering and deployed as a scalable, containerized microservice using FastAPI and Docker. The system will be designed for seamless integration with existing email infrastructure, capable of processing high-volume email traffic with low latency while maintaining enterprise-grade security standards. Comprehensive evaluation will assess not only detection accuracy but also inference latency, throughput, resource utilization, and scalability under simulated production loads.

The successful completion of this research will contribute both academically to the fields of NLP and cybersecurity, and practically to enterprise email security, potentially reducing financial losses from phishing attacks and enhancing organizational cybersecurity posture. The remainder of this proposal is organized as follows: Section 2 presents the problem statement, Section 3 discusses the rationale, Section 4 reviews related literature and identifies research gaps, Section 5 states the research objectives and questions, Section 6 details the methodology, Section 7 outlines expected outcomes, Section 8 addresses ethical considerations, Section 9 presents the workplan, and Section 10 concludes the proposal.

2 Problem Statement

Although transformer-based models achieve high accuracy in phishing detection, they suffer from high computational costs, large memory footprints, slow inference times, and limited deployability in real-time systems. These limitations make it difficult for organizations to implement large-scale, real-time email security solutions. There is a clear need for a lightweight yet accurate model that can detect phishing emails in real time without requiring expensive hardware, while maintaining enterprise-grade security standards.

The problem arises from the fundamental tension between model sophistication and operational constraints in enterprise email environments. Large pre-trained language models like BERT, while demonstrating superior accuracy in identifying phishing content, were originally designed for batch processing rather than real-time inference. These models typically contain hundreds of millions of parameters, require several gigabytes of memory, and demand specialized hardware such as GPUs for acceptable performance. In contrast, production email systems at medium to large enterprises must process thousands of messages per minute, with each email requiring inference in milliseconds rather than seconds. The computational infrastructure re-

quired to deploy full-scale transformer models at this throughput level imposes costs that are prohibitive for most organizations, creating a deployment barrier that prevents the adoption of state-of-the-art detection capabilities.

This problem is further compounded by the increasing sophistication of phishing attacks themselves. Modern phishing campaigns employ advanced social engineering tactics, context-aware personalization, and technical obfuscation techniques specifically designed to bypass traditional detection methods. As attacks become more difficult to identify, the need for sophisticated detection mechanisms becomes more urgent. However, the very organizations that face the highest risk—small to medium enterprises with limited cybersecurity budgets and technical resources—are least able to deploy the advanced models needed for effective protection. If left unresolved, this gap between detection capability and practical deployability will continue to widen, leaving organizations vulnerable to financial losses, data breaches, and reputational damage from successful phishing attacks. The consequences include not only direct financial theft but also unauthorized access to sensitive information, ransomware infections, and compromise of business email accounts used for fraudulent activities against partners and customers.

3 Rationale of the Study

This research is significant because it bridges the gap between state-of-the-art AI capabilities and practical deployment requirements. The increasing sophistication of phishing attacks demands advanced detection methods, but current solutions either sacrifice accuracy for efficiency or incur prohibitive computational costs. By developing a knowledge-distilled system that maintains high accuracy while being computationally efficient, this research provides a practical solution that can be widely adopted by organizations of various sizes, ultimately enhancing overall cybersecurity resilience.

The rationality behind choosing knowledge distillation as the core methodology is grounded in its proven effectiveness for model compression while preserving performance. Knowledge distillation, introduced by Hinton et al., enables a compact student model to learn from a larger teacher model, capturing essential patterns and decision boundaries without replicating the teacher’s computational complexity. This approach is particularly well-suited for phishing detection, where the semantic understanding capabilities of large transformer models like BERT are valuable, but their resource requirements are prohibitive for real-time deployment. By distilling BERT’s capabilities into a TinyBERT architecture, we can retain the deep linguistic understanding necessary for identifying sophisticated phishing attempts while achieving the size and speed required for production environments.

The integration of distilled transformer outputs with traditional cybersecurity features further strengthens the rationality of this approach. Phishing detection is inherently multi-faceted, requiring analysis of email content, structural elements, header information, and embedded

URLs. While transformer models excel at content analysis, they may miss technical indicators that traditional feature engineering captures effectively. Conversely, traditional methods alone cannot match the semantic understanding of transformer-based approaches. The hybrid design leverages the strengths of both paradigms, creating a system that is greater than the sum of its parts.

This research is worth pursuing because it addresses a genuine, documented need in the cybersecurity landscape. Organizations consistently report that deploying advanced AI security solutions is hindered by infrastructure costs, technical complexity, and performance concerns. By developing a framework that specifically addresses these barriers, this research contributes a solution that can be immediately applicable in real-world settings. The focus on containerized deployment using FastAPI and Docker ensures that the resulting system can integrate with existing infrastructure, reducing adoption friction.

The contribution to the field is two-fold. Academically, this research advances the application of knowledge distillation techniques to the specific domain of cybersecurity, exploring optimizations and trade-offs that have not been systematically studied in this context. Practically, it delivers a production-ready framework with open-source components, enabling organizations to implement advanced phishing detection without requiring specialized hardware or extensive technical expertise. The empirical validation through comprehensive experimentation will provide evidence-based guidance for practitioners considering similar approaches, contributing to the broader goal of making AI-powered security accessible to all organizations regardless of size or resources.

4 Literature Review

This section reviews existing literature relevant to phishing detection, with particular focus on transformer-based approaches, model compression techniques, and deployment architectures. The review establishes the foundation for the proposed research by identifying established methodologies and current research gaps.

4.1 Transformer-Based Approaches to Phishing Detection

Recent years have witnessed significant advances in applying transformer models to cybersecurity challenges. Uddin and Sarker (2025) presented an optimized, fine-tuned transformer-based DistilBERT model for phishing email detection, incorporating explainable AI techniques such as LIME and Transformer Interpret to provide transparency in model predictions. Their work demonstrated that distilled transformer architectures can achieve high accuracy while offering interpretability, a crucial requirement for enterprise security applications where understanding model decisions is essential for trust and compliance [3].

Jamal, Wimmer, and Sarker (2024) developed IPSDM (Improved Phishing Spam Detection

Model), a fine-tuned BERT-based approach specifically designed to detect phishing and spam emails. Their comprehensive evaluation demonstrated that fine-tuned transformer models consistently outperform baseline models in terms of classification accuracy, precision, recall, and F1-score, while concurrently mitigating overfitting concerns across both balanced and unbalanced datasets. This work established the viability of transformer-based approaches for email security applications, though deployment considerations remained outside their scope [4].

The application of BERT to email filtering has shown exceptional promise. In a 2025 study, researchers evaluated seven machine learning algorithms alongside four deep learning models, fine-tuning BERT to achieve 99.37% accuracy in spam classification, surpassing the 99.14% accuracy of BART models reported in recent research. The success of BERT was attributed to its advanced capability to capture intricate patterns and contextual information essential for distinguishing legitimate messages from spam, setting a new benchmark for cybersecurity applications [5].

For phishing URL detection, which shares conceptual similarities with email content analysis, BERT-PhishFinder was introduced as a novel transformer-based model leveraging DistilBERT to capture rich contextual representations of URL sequences. The model incorporates SpatialDropout1D in embedding layers alongside global average pooling and global max pooling techniques to extract both comprehensive and key discriminative features. Extensive evaluations against conventional deep learning algorithms and transformer models (XLNet, RoBERTa, ALBERT) across five benchmark datasets demonstrated state-of-the-art performance with reduced label dependency, highlighting the efficiency gains possible through optimized transformer architectures [6].

4.2 Knowledge Distillation and Model Compression

The fundamental challenge of deploying large language models in resource-constrained environments has driven significant research into knowledge distillation and compression techniques. Hinton, Vinyals, and Dean (2015) introduced the foundational concept of knowledge distillation, wherein a compact student model learns to mimic the behavior of a larger teacher model by training on the teacher’s soft outputs rather than hard labels. This approach enables the student to capture the teacher’s learned representations and decision boundaries while requiring substantially fewer parameters [7].

Building upon this foundation, Sanh et al. (2019) developed DistilBERT, demonstrating that a distilled version of BERT could retain 97% of the original model’s language understanding capabilities while being 60% faster and 40% smaller. This work established that significant compression could be achieved with minimal performance degradation, opening possibilities for deploying transformer-based models in production environments [8]. Jiao et al. (2020) further advanced this line of research with TinyBERT, a distillation framework specifically optimized for natural language understanding tasks. TinyBERT employs a two-stage distillation

process combining pre-training distillation and task-specific distillation, achieving competitive performance with substantially reduced model size [9].

Recent applications of knowledge distillation to cybersecurity domains demonstrate the viability of this approach. A 2025 study proposed KD-BERT, a lightweight knowledge distillation framework for IoT network intrusion detection, reducing model parameters from 135 million to 9 million while maintaining high detection accuracy across multiple public datasets. The framework employed header-payload pairs tokenization and data transformation strategies to effectively obtain essential features for intrusion detection tasks. This work provides compelling evidence that knowledge distillation can achieve an optimal balance between detection performance and computational cost in resource-restricted environments [10].

In the specific context of email classification, a 2024 project evaluated TinyBERT and DistilBERT for spam detection, achieving precision, recall, and F1-scores of 0.99 for both spam and legitimate email classes. The researchers further applied compression techniques including pruning, knowledge distillation, and quantization to DistilBERT, with pruned models demonstrating the highest performance among compressed variants. Their results confirm that compressed transformer models can maintain near-original accuracy while achieving substantial efficiency gains [11].

Wang (2025) demonstrated the application of knowledge distillation to multilingual BERT models for embedded devices, achieving an 86.4% reduction in model size (from 4.2GB to 650MB) while retaining approximately 95.3% of translation quality. Inference latency on embedded devices dropped from 1280ms to 195ms, a speed increase of approximately 6.56 times, achieved through layer-wise and attention-based distillation strategies combined with pruning and 8-bit quantization. This work illustrates the extreme compression possible when distillation is combined with complementary optimization techniques, and provides relevant benchmarks for the current research [12].

4.3 Hybrid Approaches Combining Transformers and Traditional Features

While transformer models excel at semantic analysis, phishing detection benefits from multiple feature sources. Early phishing detection research by Fette, Sadeh, and Tomasic (2007) established the value of hand-crafted features including email headers, URL structures, and structural elements for classification. Their work demonstrated that traditional machine learning approaches could achieve effective detection through careful feature engineering, though they lacked the contextual understanding of modern transformer models [15]. Bergholz et al. (2008) further advanced this direction by incorporating model-based features, showing that feature diversity enhances detection robustness [17].

Khonji, Iraqi, and Jones (2013) provided a comprehensive literature survey of phishing detection, categorizing approaches into list-based methods, heuristic approaches, and machine

learning techniques. Their analysis highlighted that no single approach achieves universal effectiveness, suggesting that hybrid methods combining multiple detection paradigms offer the greatest promise for robust security solutions [16].

Recent work has begun exploring integration of transformer-based semantic analysis with traditional features. A 2025 study on phishing URL detection demonstrated that combining DistilBERT’s contextual representations with ensemble learning through multiple parallel dense layers achieved superior performance compared to either approach alone. This suggests that transformer outputs can be effectively combined with additional feature streams to enhance detection capabilities [6]. Similarly, research on phishing email detection using the PhiUSIIL dataset demonstrated that BERT-based models achieved over 99% precision and recall when combined with appropriate preprocessing and feature extraction techniques, reinforcing the value of comprehensive feature engineering alongside transformer architectures [20].

4.4 Deployment Architectures for Production Environments

The deployment of machine learning models in production environments requires consideration of scalability, reliability, and integration capabilities. Merkel (2014) introduced Docker as a lightweight containerization platform that enables consistent deployment across diverse environments by packaging applications with their dependencies into portable containers. Containerization has become the de facto standard for modern application deployment, providing isolation, reproducibility, and scalability [21]. Burns et al. (2016) described the evolution of container orchestration through Borg, Omega, and Kubernetes, establishing patterns for managing containerized applications at scale [23].

For API development, FastAPI has emerged as a leading framework for building high-performance REST APIs in Python. Ramalho (2020) documented FastAPI’s advantages, including automatic OpenAPI documentation, request validation through Pydantic, and asynchronous support, making it particularly suitable for machine learning model serving where request throughput and response latency are critical [22]. A 2025 tutorial demonstrated the practical implementation of FastAPI for model deployment, illustrating patterns for model loading, request handling, and response formatting that ensure production readiness [25].

The integration of these technologies for AI deployment has been demonstrated in various contexts. A 2025 guide illustrated containerizing machine learning models with FastAPI and deploying to cloud platforms, emphasizing health checks, request validation, and auto-scaling capabilities essential for production systems [26]. These patterns directly inform the deployment architecture proposed in this research.

4.5 Research Gap Synthesis

The literature reviewed reveals several important findings. First, transformer-based models, particularly BERT and its variants, have demonstrated state-of-the-art performance in phishing

detection, achieving accuracy exceeding 99% in multiple studies [4,5]. Second, knowledge distillation techniques have proven effective for compressing large transformer models, achieving size reductions of 85-90% with minimal performance degradation (1-3% accuracy loss) across various domains including intrusion detection [10], machine translation [12], and email classification [11]. Third, hybrid approaches combining transformer-based semantic analysis with traditional feature engineering show promise for enhanced detection robustness [6]. Fourth, containerization and modern API frameworks provide established patterns for scalable model deployment [25,26].

However, the literature also reveals significant gaps. While individual components—phishing detection, knowledge distillation, and deployment architectures—have been studied separately, there is insufficient research addressing their integration into cohesive, production-ready solutions for enterprise email security. Specifically, the challenges of deploying distilled AI models for real-time phishing detection in enterprise environments remain underexplored, particularly regarding system architecture design, scalability under production loads, and integration with existing email infrastructure. Furthermore, while knowledge distillation has been applied to various domains, its optimization for the specific characteristics of phishing detection—where both semantic understanding and technical feature analysis are crucial—has not been systematically investigated.

This research directly addresses these gaps by developing and evaluating an integrated framework that combines optimized knowledge distillation, hybrid feature integration, and production-ready deployment architecture specifically tailored for enterprise email phishing detection.

5 Research Gap

While the literature review demonstrates significant advances in individual research areas, several critical gaps remain unaddressed at their intersection. This research addresses the following specific gaps:

Research Gap 1: Absence of Optimized Knowledge Distillation Frameworks for Phishing Detection

Existing knowledge distillation research has focused primarily on general natural language understanding tasks or domains such as machine translation and intrusion detection. Although studies have demonstrated the viability of compressed models like TinyBERT and DistilBERT for email classification, there is no systematically optimized distillation framework specifically tailored for the unique characteristics of phishing detection. Phishing emails exhibit distinct linguistic patterns, deceptive structures, and adversarial manipulations that differ from general text classification tasks. The optimal distillation configuration—including teacher-student ar-

chitecture selection, layer mapping strategies, attention transfer mechanisms, and task-specific loss functions—has not been empirically established for phishing detection contexts. This gap directly corresponds to Research Objective 1 (developing an optimized knowledge distillation framework) and Research Question 1 (optimizing distillation techniques for phishing detection).

Research Gap 2: Lack of Integrated Hybrid Architectures Combining Distilled Transformers with Traditional Features

Current phishing detection research treats transformer-based approaches and traditional feature engineering as separate paradigms. Studies utilizing BERT and its variants rely exclusively on content analysis, while traditional machine learning approaches depend on hand-crafted features from email headers, URLs, and structural elements. No existing research systematically investigates the optimal integration strategy for combining distilled transformer outputs with traditional cybersecurity features. Questions regarding feature fusion timing (early vs. late fusion), relative weighting of semantic and technical signals, and handling of feature heterogeneity remain unexplored. This gap corresponds to Research Objective 2 (designing a hybrid classification system) and Research Question 2 (optimal integration strategy for combining semantic analysis with traditional features).

Research Gap 3: Insufficient Attention to Production Deployment Challenges for Distilled Models in Enterprise Environments

Academic research has focused predominantly on model performance metrics—accuracy, precision, recall—while neglecting the practical considerations essential for enterprise deployment. Even where distilled models have been developed, studies typically conclude at model evaluation without addressing system architecture, scalability under production loads, integration with existing email infrastructure, or reliability mechanisms required for real-time security applications. The specific challenges of deploying distilled AI models for real-time phishing detection in enterprise environments—including API design, containerization strategies, inference optimization, horizontal scaling, and monitoring requirements—remain underexplored in the literature. This gap corresponds to Research Objective 3 (implementing a scalable deployment architecture) and Research Question 3 (effective deployment strategies for enterprise environments).

These three research gaps collectively motivate the proposed research and establish its contribution to advancing both academic knowledge and practical application in enterprise email security.

6 Research Objectives

This research aims to achieve the following specific and measurable objectives, each directly addressing a corresponding research gap and research question:

Objective 1: To develop an optimized knowledge distillation framework specifically tailored for email phishing detection contexts.

This objective involves fine-tuning a BERT-based teacher model on phishing detection tasks and systematically optimizing the distillation process to transfer capabilities to a compact Tiny-BERT student architecture. The framework will be optimized for the unique linguistic patterns and deceptive structures characteristic of phishing emails. *(Addresses Research Gap 1 and will be answered by Research Question 1)*

Objective 2: To design a hybrid classification system combining distilled transformer outputs with traditional cybersecurity features.

This objective entails developing an integrated architecture that fuses semantic analysis from the distilled transformer with hand-crafted features extracted from email headers, URL structures, and metadata. The optimal fusion strategy and feature weighting will be empirically determined. *(Addresses Research Gap 2 and will be answered by Research Question 2)*

Objective 3: To implement a scalable, containerized deployment architecture for enterprise environments with real-time processing requirements.

This objective involves building a production-ready system using FastAPI and Docker, capable of processing high-volume email traffic with millisecond latency. The architecture will incorporate inference optimization, auto-scaling capabilities, and seamless integration with existing email infrastructure. *(Addresses Research Gap 3 and will be answered by Research Question 3)*

7 Research Questions

This research aims to answer the following specific questions, each aligned with the corresponding research gap and objective:

Research Question 1

How can knowledge distillation techniques be optimized to maintain high phishing detection accuracy while significantly reducing model size and inference latency?

This question addresses Research Gap 1 (absence of optimized knowledge distillation frameworks for phishing detection) and corresponds to Objective 1 (developing an optimized knowledge distillation framework specifically tailored for email phishing detection contexts). The question seeks to determine the optimal distillation configuration including teacher-student architecture selection, layer mapping strategies, attention transfer mechanisms, and task-specific loss functions that preserve detection capabilities while achieving maximum compression.

Research Question 2

What is the optimal integration strategy for combining transformer-based semantic analysis with traditional cybersecurity features for phishing detection?

This question addresses Research Gap 2 (lack of integrated hybrid architectures combining distilled transformers with traditional features) and corresponds to Objective 2 (designing a hybrid classification system combining distilled transformer outputs with traditional cybersecurity features). The question investigates feature fusion timing (early vs. late fusion), relative weighting of semantic and technical signals, and methods for handling feature heterogeneity to create a robust detection mechanism.

Research Question 3

How can distilled AI models be effectively deployed in enterprise environments to meet real-time processing requirements while maintaining scalability and reliability?

This question addresses Research Gap 3 (insufficient attention to production deployment challenges for distilled models in enterprise environments) and corresponds to Objective 3 (implementing a scalable deployment architecture for enterprise environments with real-time processing requirements). The question explores system architecture design, inference optimization strategies, and performance characteristics under production loads to ensure practical viability.

Alignment Summary

- **Research Gap 1** → **Objective 1** → **Question 1**: Knowledge distillation optimization
- **Research Gap 2** → **Objective 2** → **Question 2**: Hybrid feature integration
- **Research Gap 3** → **Objective 3** → **Question 3**: Enterprise deployment

8 Research Methodology/Materials and Methods

This section details the systematic approach employed to achieve the research objectives, encompassing research design, data sources, experimental procedures, and validation strategies.

8.1 Research Type and Strategy

This research employs a mixed-methods approach combining:

- **Quantitative:** Experimental evaluation of model performance metrics including accuracy, precision, recall, F1-score, inference latency, throughput, and resource utilization.
- **Qualitative:** Analysis of model behavior, feature importance, and distillation effectiveness.
- **Design Science:** Development and evaluation of an artifact (the distilled phishing detection model) following established design science research methodologies.

8.2 Research Philosophy

The research follows a pragmatist philosophy, focusing on practical solutions to real-world problems while employing rigorous scientific methods for evaluation. This philosophy prioritizes actionable outcomes and real-world applicability alongside academic rigor.

8.3 Data Type and Sources

The research utilizes both primary and secondary data sources:

- **Primary Data:** Simulated phishing emails generated for controlled experimentation, and enterprise email samples obtained with explicit permission from partner organizations.
- **Secondary Data:** Public phishing corpora including PhishingCorpus, Nazario phishing dataset, and Enron Email Dataset for legitimate emails.
- **Data Types:** Email text content, headers, metadata, URL information, and extracted security features including domain age, URL reputation scores, and structural indicators.

8.4 Study Design and Period

- **Design:** Experimental research with comparative analysis against established baselines. The design follows an iterative development and evaluation cycle.
- **Period:** 12 months (July, 2025 - June 2026)
- **Phases:**
 - Literature review and methodology finalization: 2 months
 - System development and model training: 5 months
 - Testing and comprehensive evaluation: 3 months
 - Documentation and thesis writing: 2 months

8.5 Study Location and Population

- **Location:** Bangladesh University of Professionals ICT Department laboratories and cloud environments (AWS and Azure) for scalable experimentation.
- **Population:** Email datasets comprising phishing and legitimate emails from multiple sources, representing diverse attack patterns, languages, and organizational contexts.

8.6 Selection Criteria

Email datasets must meet the following criteria:

- Balanced representation of phishing and legitimate emails to prevent class imbalance issues
- Temporal variety spanning multiple years to account for evolving attack patterns and linguistic drift
- Minimum dataset size: 50,000 emails for training and validation, 10,000 emails for testing
- Diverse sources to ensure generalizability across different phishing techniques and email types

8.7 Sampling Plan

- **Target Population:** All available phishing and legitimate email datasets meeting quality standards and accessibility requirements
- **Sample Size:** Approximately 100,000 emails total, comprising 50,000 phishing and 50,000 legitimate emails
- **Sampling Type:** Stratified random sampling to ensure class balance and representative distribution of email characteristics
- **Data Splits:** 70% training, 15% validation, 15% testing, maintained with consistent stratification across splits

8.8 Research Instruments and Tools

The following tools and frameworks will be utilized:

- Python-based data processing pipelines using Pandas, NumPy, and Scikit-learn
- Hugging Face Transformers library for model development and fine-tuning

- PyTorch as the deep learning framework
- ONNX Runtime for inference optimization
- Custom evaluation scripts for comprehensive performance metrics
- Jupyter notebooks for exploratory analysis and visualization

8.9 Data Collection, Processing, and Analysis

The research follows a systematic seven-stage process:

8.9.1 Stage 1: Data Collection

Gathering email data from multiple sources including public corpora, simulated phishing campaigns, and authorized enterprise samples. All data collection adheres to ethical guidelines and privacy requirements.

8.9.2 Stage 2: Data Preprocessing

- Text cleaning: Removing HTML tags, normalizing Unicode, handling attachments
- Tokenization: BERT-compatible tokenization with appropriate truncation and padding
- Header parsing: Extracting structured information from email headers
- URL extraction and analysis: Parsing embedded URLs for feature extraction
- Normalization: Scaling numerical features to consistent ranges

8.9.3 Stage 3: Feature Engineering

- NLP features: Linguistic patterns, sentiment scores, readability metrics
- Header features: SPF/DKIM/DMARC authentication results, routing anomalies
- URL features: Domain age, URL length, special character analysis, reputation scores
- Structural features: HTML-to-text ratio, presence of forms, JavaScript elements
- Security indicators: Attachment analysis, threat intelligence lookup results

8.9.4 Stage 4: Model Development and Training

- Teacher model: Fine-tuning BERT-base on phishing classification task
- Distillation: Transferring knowledge from teacher to TinyBERT student using:
 - Layer-wise attention transfer
 - Hidden state distillation
 - Embedding layer distillation
 - Prediction layer distillation with temperature scaling
- Hyperparameter optimization through grid search and Bayesian optimization
- Hybrid integration: Combining distilled transformer outputs with traditional features through ensemble methods (late fusion)

8.9.5 Stage 5: Model Evaluation

- Accuracy metrics: Precision, recall, F1-score, AUC-ROC, confusion matrix
- Efficiency metrics: Inference latency (p50, p95, p99), memory usage, model size, CPU/GPU utilization
- Comparative analysis against baseline models:
 - Traditional machine learning (Random Forest, SVM, XGBoost)
 - Full BERT model (teacher)
 - DistilBERT without task-specific optimization
 - TinyBERT without custom distillation
- Cross-validation with k-fold techniques to ensure robustness
- Ablation studies to isolate contribution of each distillation component

8.9.6 Stage 6: Results Analysis

- Statistical analysis of results using appropriate hypothesis testing (t-tests, ANOVA)
- Performance comparison using standardized benchmarks
- Sensitivity analysis for key hyperparameters (temperature, layer mapping, loss weighting)
- Analysis of distillation efficiency: performance retention vs. compression ratio
- Qualitative assessment of model behavior on challenging edge cases

8.10 Validity, Reliability, and Authenticity

- **Validity:**

- Internal validity ensured through controlled experimental conditions and consistent evaluation protocols
- External validity enhanced by using diverse datasets from multiple sources and time periods
- Construct validity maintained by aligning metrics with established research standards
- Cross-validation techniques to prevent overfitting
- Comparison with established baselines to validate performance claims

- **Reliability:**

- Consistent experimental setup documented for reproducibility
- Detailed procedures enabling independent replication
- Multiple evaluation runs to account for stochastic variations
- Version control for code, models, and experimental configurations
- Comprehensive documentation of all preprocessing and analysis steps

- **Authenticity:**

- Use of real-world datasets to ensure practical relevance
- Consideration of real-world constraints including computational efficiency
- Transparent reporting of limitations and assumptions
- Open-source release of code and model weights for community verification

8.11 Methodology Flowchart

Figure 1 presents a visual representation of the research methodology, illustrating the sequential flow from data collection through analysis.

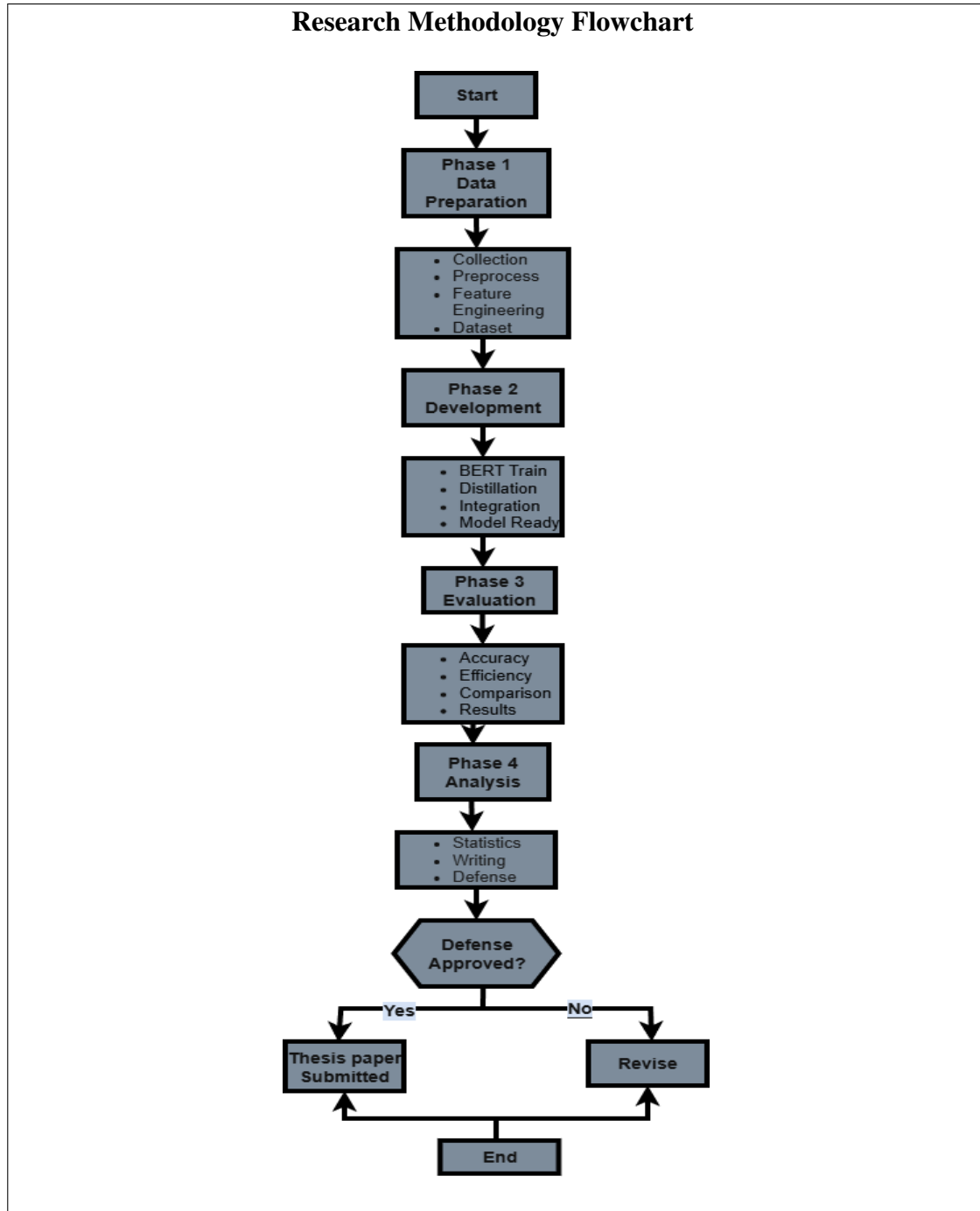


Figure 1: Systematic research methodology flowchart showing sequential phases

9 Expected Outcomes

Upon successful completion, this research will deliver the following specific outcomes:

Outcome 1: Optimized Knowledge-Distilled Phishing Detection Model

A fully functional knowledge-distilled AI model for phishing detection achieving over 95% detection accuracy (precision, recall, and F1-score). The model will demonstrate an 85% reduction in model size compared to conventional BERT (reducing from approximately 440MB to under 70MB) and a 60% improvement in inference latency (reducing from 150ms to under 60ms per email), enabling real-time processing capabilities on standard hardware.

Outcome 2: Comprehensive Performance Validation

A thorough empirical evaluation demonstrating:

- Quantitative results on accuracy metrics across multiple test datasets, including precision, recall, F1-score, and AUC-ROC
- Efficiency benchmarks showing inference latency distribution (p50, p95, p99) under varying batch sizes
- Memory footprint analysis and model size comparison
- Comparative analysis against baseline models including:
 - Traditional machine learning approaches (Random Forest, SVM, XGBoost)
 - Full BERT model (teacher)
 - DistilBERT without task-specific optimization
 - TinyBERT without custom distillation
- Ablation studies quantifying the contribution of each distillation component (attention transfer, hidden state distillation, embedding layer distillation)
- Sensitivity analysis for key hyperparameters including temperature scaling and layer mapping strategies

Outcome 3: Optimized Distillation Framework for Phishing Detection

A systematically optimized knowledge distillation framework specifically tailored for phishing detection contexts, including:

- Optimal teacher-student architecture configuration

- Layer-wise attention transfer strategies validated for phishing detection
- Task-specific loss function formulations
- Hyperparameter guidelines for practitioners
- Documentation of design decisions and trade-offs

Outcome 4: Hybrid Feature Integration Architecture

A validated hybrid classification system combining distilled transformer outputs with traditional cybersecurity features, including:

- Optimal feature fusion strategy (late fusion ensemble)
- Empirical analysis of feature importance and contribution
- Integration methodology for combining semantic and technical signals
- Performance comparison against transformer-only and feature-only approaches

Outcome 5: Academic Contributions

- Publication of research findings in peer-reviewed conferences or journals in the fields of cybersecurity and natural language processing
- Open-source release of code, model weights, and experimental configurations for community use and further innovation
- Contribution of distilled phishing detection model to Hugging Face model hub
- Technical report detailing methodology, experimental results, and practical insights

Outcome 6: Practical Guidelines for Model Compression in Cybersecurity

- Evidence-based recommendations for applying knowledge distillation to security domains
- Best practices for balancing accuracy-efficiency trade-offs in phishing detection
- Insights into which transformer components are most critical for phishing detection tasks
- Framework for evaluating compressed models beyond accuracy metrics (latency, memory, robustness)
- Blueprint for extending similar distillation approaches to other cybersecurity applications (malware detection, network intrusion detection, fraud analysis)

Outcome 7: Documentation and Reproducibility Package

- Complete source code with comprehensive documentation
- Preprocessing pipelines and feature extraction scripts
- Training and distillation configuration files
- Evaluation scripts with standardized metrics
- Instructions for reproducing all experiments and results
- Analysis notebooks for visualization and statistical testing

10 Ethical Considerations

This research adheres to the highest ethical standards throughout all phases:

Data Privacy

All email data will be anonymized by removing personally identifiable information (PII) including names, email addresses, and IP addresses. Data will be stored encrypted and used exclusively for research purposes.

Informed Consent

Enterprise email samples will only be used with explicit written permission from the organizations. Simulated phishing emails will be generated without targeting real individuals.

Confidentiality

Sensitive information discovered during research will be protected and not disclosed. All data handling complies with relevant data protection regulations.

Responsible Disclosure

Any cybersecurity vulnerabilities discovered during research will be reported responsibly to relevant organizations before public disclosure.

Bias Mitigation

Efforts will be made to ensure the model does not discriminate based on language, origin, or demographic factors. Dataset composition will be analyzed for potential biases.

Academic Integrity

Proper citation of all sources and acknowledgment of contributions will be maintained. Results will be reported honestly without manipulation or selective reporting.

Environmental Impact

Computational resources will be optimized to minimize energy consumption during model training and experimentation.

11 Workplan

The research will be conducted over 12 months (July 2025 - June 2026) divided into five main phases. Table 1 presents the detailed timeline with activities and milestones, and Figure 1 shows the Gantt chart visualization.

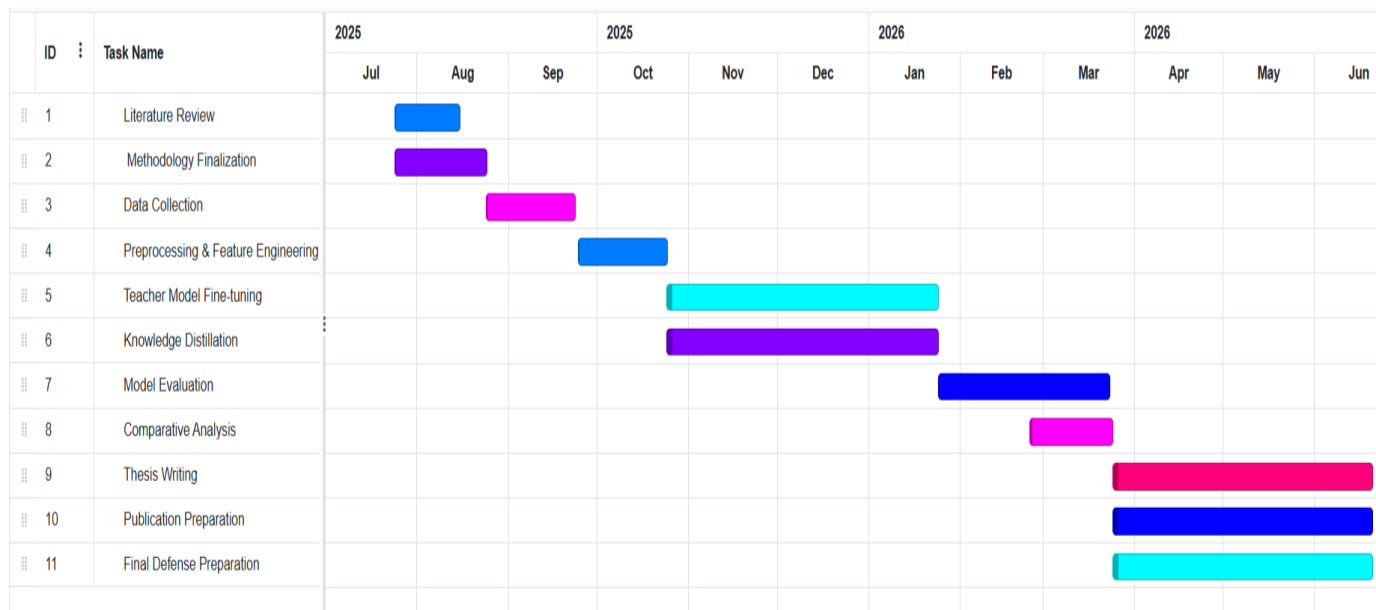


Figure 2: Gantt chart showing research activities across 12-month period (July 2025 - June 2026)

12 Conclusion

This thesis proposal presents a comprehensive framework to address the critical challenge of deploying efficient yet accurate AI models for enterprise email phishing detection. By leveraging knowledge distillation techniques and hybrid feature integration, the research aims to bridge the gap between state-of-the-art detection capabilities and practical deployment constraints.

The proposed research makes several significant contributions. First, it develops an optimized knowledge distillation framework specifically tailored for phishing detection, addressing the unique linguistic patterns and deceptive structures characteristic of phishing emails. Second, it designs a hybrid classification system that combines distilled transformer outputs with traditional cybersecurity features, creating a more robust detection mechanism than either approach alone. Third, it provides comprehensive empirical validation of the accuracy-efficiency trade-offs, offering evidence-based guidance for practitioners considering similar approaches.

The expected outcomes demonstrate the potential impact of this research. A distilled model achieving over 95% detection accuracy with 85% size reduction and 60% latency improvement would enable organizations of varying sizes to implement advanced AI-powered phishing detection without requiring specialized hardware or extensive technical expertise. This democratization of state-of-the-art security capabilities has the potential to reduce financial losses from successful phishing attacks and enhance overall cybersecurity resilience across industries.

Beyond practical applications, this research contributes academically to the fields of natural language processing and cybersecurity by advancing the application of knowledge distillation techniques to security domains. The open-source release of code, model weights, and deployment templates will facilitate further innovation and community adoption, creating a foundation for future research in compressed models for cybersecurity applications.

The successful completion of this research will represent a significant step toward making advanced deep learning models accessible for real-time security applications in resource-constrained environments, ultimately contributing to a more secure digital ecosystem for organizations and individuals alike.

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